sota.id Integration Guide

> **Last updated:** 2026-03-06 **Swagger UI:** https://sota.id/api/public-schema/swagger-ui/ **OpenAPI schema:** https://sota.id/api/public-schema/?format=json **Base URL:** `https://sota.id/api`

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1. Overview

sota.id is the Kazakh football data aggregation platform that serves as QFL's primary source of truth for all match data, player statistics, and live events. QFL pulls data from sota.id via a scheduled Celery sync pipeline and stores it in PostgreSQL.

Data Flow

```
sota.id API
    │
    ▼
SotaClient (httpx, JWT, retry)
    │
    ▼
SyncOrchestrator
    ├── ReferenceSyncService  → tournaments, seasons, teams
    ├── PlayerSyncService     → players, player season stats
    ├── GameSyncService       → games, game events, game stats
    ├── LineupSyncService     → pre-game lineups, live positions
    └── StatsSyncService      → score table, team season stats
    │
    ▼
PostgreSQL
    │
    ▼
```

```
FastAPI → Frontend (Next.js)
```

Key Files

File	Role
`backend/app/services/sota_client.py`	HTTP client — all API calls to sota.id
`backend/app/services/sync/orchestrator.py`	Sync pipeline coordinator
`backend/app/services/sync/reference_sync.py`	Tournaments, seasons, teams
`backend/app/services/sync/player_sync.py`	Players and player stats
`backend/app/services/sync/game_sync.py`	Games, events, formations
`backend/app/services/sync/lineup_sync.py`	Pre-game + live lineups
`backend/app/services/sync/stats_sync.py`	Score table, team season stats
`backend/app/services/sync/base.py`	Shared constants (stats field sets)
`backend/app/tasks/__init__.py`	Celery Beat schedule
`backend/app/utils/lineup_feed_parser.py`	Normalize `/em/` and VSporte lineup entries
`scripts/resync_assists.py`	Manual assist re-linking tool

Implemented vs Not Implemented (summary)

- **Implemented:** ~17 of 31 endpoints (references, players, games, stats, live /em/)
- **Not implemented:** ~14 endpoints (vmix TV-broadcast, best_players, team_of_week, per-tour stats, player game stats v2)
- **CRITICAL:** Several breaking API changes detected 2026-03-06 — see [SOTA-5] in backlog

---

2. Authentication

Obtaining a Token

```
POST https://sota.id/api/auth/token/
Content-Type: application/json

{
  "email": "...",
  "password": "..."
}
```

Response:

```
{
  "access": "eyJhbGci...",
  "refresh": "eyJhbGci...",
  "multi_token": "eyJhbGci..."
}
```

Field	Purpose

`access`	Bearer token for REST API calls (TTL ~23h)
`refresh`	Token refresh (not currently used by QFL)
`multi_token`	Used as `access_token=` query param on `/em/` live endpoints

Using the Token

REST API endpoints — Authorization header:

```
Authorization: Bearer <access_token>
Accept-Language: ru    # Optional: ru | kk | en
```

Live `/em/` endpoints — Query parameter:

```
https://sota.id/em/{game_sota_id}-list.json?access_token=<access_token>
```

Retry Policy (SotaClient)

```
@retry(
    stop=stop_after_attempt(3),
    wait=wait_exponential(multiplier=1, min=2, max=8),  # 2s → 4s → 8s
    retry=retry_if_exception_type((
        httpx.ConnectTimeout,
        httpx.ReadTimeout,
        httpx.ConnectError,
        httpx.RemoteProtocolError,
    ))
)
```

The `ensure_authenticated()` method is called before every request and refreshes the JWT if expired.

---

3. Endpoint Catalog

> Tested against **Season 2025 Premier League** (`season_id=61`, `tournament_id=7`) and **Season 2026** (`season_id=173`) All responses from real curl tests on 2026-03-01, updated 2026-03-06
>
> **API Breaking Changes (2026-03-06):** Seasons/Teams/Games endpoints switched from paginated to flat list responses. Teams endpoint lost `logo`, `city`, `short_name` fields. Season names are now just years. `pass_accuracy` removed from game stats. Players `team` object became `teams` array of IDs. See [SOTA-5] in backlog.

3.1 Registry (Reference Data)

Method	Endpoint	QFL Status	Description
GET	`/api/public/v1/tournaments/`	✓ Implemented	List of tournaments
GET	`/api/public/v1/seasons/`	✓ Implemented	List of seasons (paginated, 3 languages)
GET	`/api/public/v1/teams/?season_id={id}`	✓ Implemented	Teams in a season

GET	`/api/public/v1/players/?season_id={id}`	✓ Implemented	Players in a season
GET	`/api/public/v1/metrics/`	✗ Not used	Metrics registry (293 total)

GET /api/public/v1/tournaments/

```
[
  {"id": 7, "name": "Premier League"},
  {"id": 30, "name": "First League"},
  {"id": 65, "name": "Cup. Main tournament"},
  {"id": 74, "name": "Second League Northeast Conference"},
  {"id": 75, "name": "Second League Southwest Conference"},
  {"id": 78, "name": "Womens League"},
  {"id": 139, "name": "Second League Final"}
]
```

> **Changed (2026-03-06):** Tournament IDs changed (was 1,2,3 — now 7,30,65...). `short_name` removed. Names in English regardless of `Accept-Language`.

GET /api/public/v1/seasons/

Response is now a **flat list** (not paginated):

```
[
  {
    "id": 173,
    "name": "2026",
    "start_date": "2026-03-06",
    "end_date": "2026-11-22",
    "tournament_id": 7,
    "tournament_name": "Premier League"
  },
  {
    "id": 61,
    "name": "2025",
    "start_date": "2025-03-01",
    "end_date": "2025-10-27",
    "tournament_id": 7,
    "tournament_name": "Premier League"
  }
]
```

> **Changed (2026-03-06):** - Response switched from paginated `{count, results}` to flat list - `tournament` (int) → `tournament_id` + `tournament_name` - `date_start/date_end` → `start_date/end_date` (field names changed!) - `name` is now just the year ("2026"), not full tournament+year name - `Accept-Language` has no effect on season names - `tours_count` removed

GET /api/public/v1/teams/?season_id=173

Response is now a **flat list** with nested tournament/season data:

```
[
  {
    "id": 51,
    "name": "Aktobe",
    "tournaments": [
      {
        "id": 7,
```

```
        "name": "Premier League",
        "seasons": [{"id": 10, "name": "2024"}, {"id": 61, "name": "2025"}, {"id": 173,
      },
      {
        "id": 65,
        "name": "Cup. Main tournament",
        "seasons": [{"id": 71, "name": "2025"}]
      }
    ],
    "seasons": [10, 61, 71, 173]
  }
]
```


> **Changed (2026-03-06):** - Response switched from paginated to flat list - Removed: `short_name`, `country`, `logo`, `stadium`, `city` - Added: `tournaments` (nested array with seasons), `seasons` (flat ID array) - `Accept-Language` still works for `name` (ru=Актобе, kk=Ақтөбе, en=Aktobe) - KPL 2026 has 16 teams (was 12 in 2025)

GET /api/public/v1/players/?season_id=173

Still paginated (`count`, `results`, `next`), page size = 10:


```
{
  "count": 332,
  "next": "http://sota.id/api/public/v1/players/?limit=5&offset=0&page=2&season_id=173",
  "results": [
    {
      "id": "01dc336a-47c0-4b24-8a62-8574ef990cb3",
      "first_name": "Неманья",
      "last_name": "Кавнич",
      "teams": [90],
      "birthday": "1995-09-05",
      "type": "defence",
      "country_name": "Черногория",
      "age": 30,
      "top_role": "ЦЗ (центральный защитник)"
    }
  ]
}
```


> **Changed (2026-03-06):** - `team` (object `{id, name}`) → `teams` (array of team IDs like `[90]`) - Added: `birthday`, `type` (position type: "defence"/"offence"/"midfield"/"goalkeeper") - `country` → `country_name` - `position` (single char D/M/F/G) → `type` (full word) - `top_role` now in Russian when `Accept-Language: ru` - `next` URL uses `http://` (not https) — need to handle in pagination - Names affected by `Accept-Language` (ru=Неманья Кавнич, en=Nemanja Cavnic)

GET /api/public/v1/metrics/

Returns full registry of 293 statistical metrics:


```
{
  "result": "Success",
  "data": [
    {"id": 1, "name": "Голы", "key": "goal"},
    {"id": 2, "name": "Голевые передачи", "key": "goal_pass"},
    {"id": 3, "name": "Удары", "key": "shot"}
  ]
}
```

3.2 Games

Method	Endpoint	QFL Status	Description
GET	`/api/public/v1/games/?season_id={id}&tour={n}`	⚠ Partially used	List of games (11 filter params)
GET	`/api/public/v1/games/{id}/players/`	✓ Implemented	Per-game player stats
GET	`/api/public/v1/games/{id}/teams/`	✓ Implemented	Per-game team stats
GET	`/api/public/v1/games/{id}/pre_game_lineup/`	✓ Implemented	Pre-game lineup + referees
GET	`/api/public/v1/games/{id}/vmix/generate_link/`	✗ Not used	VMix TV links + reveals undocumented /em/ endpoints
GET	`/api/public/v1/games/vmix/{id}/best-players/{team_id}/?metric={key}`	✗ Not used	Best players by metric (vmix)
GET	`/api/public/v2/games/vmix/{id}/team-players-stats/{team_id}/`	✗ Not used	All player stats matrix (vmix)

GET /api/public/v1/games/?season_id=173&tour=1

Response is now a **flat list**:

```
[
  {
    "id": "691bfa4b-...",
    "date": "2026-03-08",
    "tournament_id": 7,
    "home_team": {"id": 91, "name": "Astana", "score": 0},
    "away_team": {"id": 45, "name": "Jetisu", "score": 0},
    "tour": 1,
    "has_stats": false,
    "season_id": 173,
    "season_name": "2026",
    "visitors": null
  }
]
```

> **Changed (2026-03-06):** - Response switched from paginated to flat list - `home_score/away_score` → `home_team.score/away_team.score` (nested) - Removed: `time`, `stadium` - Added: `tournament_id`, `season_id`, `season_name`, `has_stats` - Team names in English regardless of `Accept-Language`
>
> **Filter parameters:** `season_id`, `team`, `tour`, `date_from`, `date_to`, `status`, `limit`, `offset`, `lang`

GET /api/public/v1/games/{id}/players/

```
{
  "data": {
    "latest_update_date_time": "2025-03-08T18:00:00Z",
    "players": [
      {
        "id": "01dc336a-47c0-4b24-8a62-8574ef990cb3",
        "first_name": "Nemanja",
        "last_name": "Cavnic",
        "team": "Кайсар",
        "team_id": 94,
```

```
        "minutes_played": 90,
        "started": true,
        "position": "M",
        "stats": {
          "shot": 2,
          "shots_on_goal": 1,
          "shots_off_goal": 1,
          "pass": 45,
          "pass_accuracy": 88.9,
          "duel": 8,
          "tackle": 3,
          "corner": 0,
          "offside": 0,
          "foul": 2,
          "yellow_cards": 0,
          "red_cards": 0
        }
      }
    ]
  }
}
```


GET /api/public/v1/games/{id}/teams/


```
{
  "data": {
    "teams": [
      {
        "id": 94,
        "name": "Кайсар",
        "stats": {
          "possession": 55,
          "possession_percent": "55%",
          "shot": 14,
          "shots_on_goal": 6,
          "shots_off_goal": 5,
          "pass": 420,
          "pass_accuracy": 84.3,
          "foul": 12,
          "yellow_cards": 2,
          "red_cards": 0,
          "corner": 5,
          "offside": 2
        }
      }
    ]
  }
}
```


GET /api/public/v1/games/{id}/pre_game_lineup/


```
{
  "data": {
    "date": "2025-03-08",
    "referee": {
      "main": "Иван Иванов",
      "1st_assistant": "Петр Петров",
      "2nd_assistant": "Сергей Сергеев",
      "4th_referee": "Алексей Алексеев",
      "video_assistant_1": null,
      "video_assistant_main": null,
```

```
      "match_inspector": null
    },
    "home_team": {
      "id": 94,
      "name": "Кайсар",
      "short_name": "КАЙ",
      "bas_logo_path": "C:\\KPL_FINAL\\LOGO_KOMAND\\Kaisar.png",
      "coach": "Владимир Федотов",
      "first_assistant": null,
      "second_assistant": null,
      "lineup": [
        {
          "player_id": "01dc336a-47c0-4b24-8a62-8574ef990cb3",
          "first_name": "Nemanja",
          "last_name": "Cavnic",
          "player_number": 8,
          "country_name": "Сербия",
          "bas_image_path": "C:\\KPL_FINAL\\SOTA_PHOTO_PLAYER\\Kaisar\\CAVNIC_NEMANJA.pn
        }
      ]
    },
    "away_team": { "...": "same structure" }
  }
}
```


> **Note:** `data` may be `null` for games where lineup has not been entered in sota.id.

3.3 Seasons

Method	Endpoint	QFL Status	Description
GET	`/api/public/v1/seasons/{id}/score_table/`	✓ Implemented	Championship table (v1)
GET	`/api/public/v1/seasons/{id}/score_table_v2/`	✗ No data for s.61	Championship table (v2)
GET	`/api/public/v2/seasons/{id}/score_table/`	✗ No data for s.61	Championship table (v2 alt)
GET	`/api/public/v1/seasons/{id}/best_players/?metric={key}&max={n}&tour={n}`	✗ ☆☆☆ HIGH	Best players by metric (empty for past seasons!)
GET	`/api/public/v1/seasons/{id}/team_of_week/?tour={n}`	✗ ☆☆☆ HIGH	Team of the week/tour (empty for past seasons with tour param)
GET	`/api/public/v1/seasons/{id}/season_stats_v2/`	✗ Not used	71 league-wide metrics
GET	`/api/public/v2/seasons/{id}/season_stats/`	✗ Not used	Same 71 metrics (v2 alt)
GET	`/api/public/v1/seasons/vmix/{id}/last-games/{team_id}/`	✗ Not used	Last 5 games (vmix)
GET	`/api/public/v1/seasons/vmix/{id}/score_table/`	✗ Not used	Table (vmix format)
GET	`/api/public/v1/seasons/vmix/{id}/teams_season_stats/?metric={key}`	✗ Not used	Season stats ranking

GET	`/api/public/v1/seasons/vmix/{id}/teams_average_season_stats/{game_id}/`	✕ Not used	Two-team avg stats
GET	`/api/public/v1/seasons/vmix/{id}/teams/{team_id}/best-players/{game_id}/?metric={key}`	✕ Not used	Team best players

GET /api/public/v1/seasons/61/score_table/

```
{
  "result": "success",
  "data": {
    "table": [
      {
        "id": 7,
        "name": "Кайрат",
        "logo": "https://sota.id/media/...",
        "rg": 34,
        "wins": 18,
        "draws": 5,
        "losses": 3,
        "goals": "53:19",
        "points": 59,
        "matches": 26,
        "form": ["W", "W", "D", "W", "L"]
      }
    ]
  }
}
```

> **Note:** `goals` is a string `"scored:conceded"`, not separate fields. `rg` = goal difference.

GET /api/public/v1/seasons/61/best_players/?metric=goal&max=10&tour=15

```
{
  "result": "Success",
  "data": {
    "latest_update_date_time": "2026-02-18 14:34",
    "tour": 15,
    "players": []
  }
}
```

> **WARNING (2026-03-06):** Returns empty `players: []` for ALL completed seasons (61, 10), regardless of `tour` param. Data appears to be cleared after season ends. Only useful during active season — must cache in DB.

GET /api/public/v1/seasons/61/team_of_week/

Without `tour` param returns last 3 tours' squads (33 players).

```
{
  "result": "Success",
  "data": {
    "players": [
      {
        "id": "38408933-60ce-49f1-909c-9630d39fc6c7",
        "first_name": "Dinmukhamed",
        "last_name": "Karaman",
        "full_name": "Dinmukhamed Karaman",
```

```
        "team": {
          "id": 92,
          "name": "Jenis",
          "image": "https://videos.sota.id/team/image/Jenis.webp"
        },
        "number": 8,
        "captain": false,
        "amplua": "D",
        "position": "L",
        "ordering": 1
      }
    ]
  }
}
```

> **Changed (2026-03-06):** - Added: `full_name`, `number`, `captain`, `team.image` (logo URL on videos.sota.id) - `value` field removed - Without `tour` returns 33 players (last 3 tours), not single tour - With `tour=N` returns empty for completed seasons (data cleared)

```
#### GET /api/public/v1/seasons/61/season_stats_v2/

Returns 71 league-wide aggregate metrics:
```json
{
  "data": {
    "stats": [
      {"key": "goal", "name": "Голы", "value": 312},
      {"key": "shot", "name": "Удары", "value": 3420},
      {"key": "yellow_cards", "name": "Жёлтые карточки", "value": 408}
    ]
  }
}
```
```

3.4 Teams

Method	Endpoint	QFL Status	Description
GET	`/api/public/v1/teams/{id}/season_stats/?season_id={id}`	✓ Implemented	Season stats v1 (10 fields)
GET	`/api/public/v2/teams/{id}/season_stats/?season_id={id}`	✓ Implemented	Season stats v2 (92 metrics)
GET	`/api/public/v1/teams/{id}/season_stats_v2/?season_id={id}&tour={n}&single_tour={bool}`	✗ ☆☆	Per-tour stats

GET /api/public/v2/teams/94/season_stats/?season_id=61

```
{
  "data": {
    "stats": [
      {"key": "games_played", "name": "Игры", "value": 26},
      {"key": "win", "name": "Победы", "value": 8},
      {"key": "draw", "name": "Ничьи", "value": 7},
      {"key": "match_loss", "name": "Поражения", "value": 11},
      {"key": "goal", "name": "Голы", "value": 35},
      {"key": "goals_conceded", "name": "Пропущено", "value": 48},
      {"key": "shot", "name": "Удары", "value": 348},
```

```
      {"key": "pass", "name": "Передачи", "value": 9450},
      {"key": "possession_percent_average", "name": "Владение", "value": 43.2}
    ]
  }
}
```


Full response contains **92 metrics** covering: shots, passes, duels, defense, set pieces, discipline, goalkeeping.

GET /api/public/v1/teams/94/season_stats_v2/?season_id=61&tour=5&single_tour=false

Same 92-metric format but cumulative through tour 5 (or single tour if `single_tour=true`).

3.5 Players

Method	Endpoint	QFL Status	Description
GET	`/api/public/v1/players/{id}/season_stats/?season_id={id}`	⚠ Deprecated	Season stats v1 (14 fields)
GET	`/api/public/v2/players/{id}/season_stats/?season_id={id}`	✓ Implemented	Season stats v2 (50 metrics)
GET	`/api/public/v2/players/{id}/game_stats/?game_id={gid}`	✗ ☆☆☆ HIGH	Per-game stats (50 metrics)

GET /api/public/v2/players/{id}/season_stats/?season_id=61


```
{
  "data": {
    "stats": [
      {"key": "games_played", "name": "Игры", "value": 24},
      {"key": "goal", "name": "Голы", "value": 3},
      {"key": "goal_pass", "name": "Ассисты", "value": 5},
      {"key": "shot", "name": "Удары", "value": 45},
      {"key": "pass", "name": "Передачи", "value": 1240},
      {"key": "pass_ratio", "name": "Точность передач", "value": 87.3},
      {"key": "dribble", "name": "Дриблинги", "value": 28},
      {"key": "yellow_cards", "name": "Жёлтые карточки", "value": 4}
    ]
  }
}
```


GET /api/public/v2/players/{id}/game_stats/?game_id={game_uuid}

Same 50-metric structure scoped to a single game. Can also filter by `season_id` and `tour`.

GET /api/public/v1/players/{id}/season_stats/ (deprecated v1)


```
{
  "data": {
    "games_played": 24,
    "goals": 3,
    "assists": 5,
    "shots": 45,
    "passes": 1240,
    "pass_accuracy": 87.3,
    "yellow_cards": 4,
    "red_cards": 0
  }
```

```
}
```

Flat object with 14 fields. Superseded by v2.

3.6 Live Match Endpoints (`/em/`)

These endpoints are **outside the Swagger spec** and use `?access_token=` auth (not Authorization header).

Endpoint	QFL Status	Description
`/em/{sota_id}-team-home.json`	✓ Implemented	Home team live lineup
`/em/{sota_id}-team-away.json`	✓ Implemented	Away team live lineup
`/em/{sota_id}-list.json`	✓ Implemented	Match events (goals, cards, subs)
`/em/{sota_id}-stat.json`	✓ Implemented	Live match stats (109 metrics)
`/em/{sota_id}-referies.json`	✓ Read by vmix	Referee assignments
`/em/{sota_id}-lineups-home.json`	✗ Not used	Simplified home lineup (number+name)
`/em/{sota_id}-lineups-away.json`	✗ Not used	Simplified away lineup
`/em/{sota_id}-players-home.json`	✗ Not used	Home player stats + goals/assists/cards
`/em/{sota_id}-players-away.json`	✗ Not used	Away player stats + goals/assists/cards

> **Discovery:** The `/em/` endpoint list was found via `GET /api/public/v1/games/{id}/vmix/generate_link/` which returns all available stream links including undocumented `/em/` variants.

`/em/{id}-team-home.json` — Live Lineup


```
[
  {"number": "TEAM",       "first_name": "Жетісу", "last_name": "home", "full_name": "C:\
  {"number": "FORMATION", "first_name": "4-3-3 down", "last_name": "ЖЕТ", "full_name": "
  {"number": "COACH",      "first_name": "Самат", "last_name": "Смаков", "full_name": "Ca
  {"number": "MAIN",       "first_name": "Самат", "last_name": "Смаков", "full_name": "Ca
  {"number": "ОСНОВНЫЕ",   "first_name": "", "last_name": "", "full_name": "", "gk": "",
  {
    "number": 31,
    "first_name": "Михаил",
    "last_name": "Голубничий",
    "full_name": "Михаил Голубничий",
    "gk": true,
    "capitan": false,
    "bas_image_path": "C:\\KPL_FINAL\\SOTA_PHOTO_PLAYER\\Zhetysu\\MIHAIL_GOLUBNICHII.png
    "amplua": "Gk",
    "position": "C",
    "id": "205d6bbe-002d-452b-8539-4935084b6371"
  },
  {"number": "ЗАПАСНЫЕ", "...": "marker row — players below are substitutes"},
  {"number": 25, "first_name": "...", "...": "substitute player"}
]
```


Marker tokens:

Token	Meaning
`TEAM`	Team name + side ("home"/"away"), logo BAS path in `full_name`

FORMATION	Formation string (e.g. "4-3-3 down"), team short name, kit color #RRGGBB in full_name
COACH	Coach name (appears as both COACH and MAIN entries)
MAIN	Primary coach marker
ОСНОВНЫЕ	"STARTERS" section begins — players below are starters
ЗАПАСНЫЕ	"SUBSTITUTES" section begins — players below are subs
STADIUM/VENUE	Stadium name in first_name
TIME/DATE	Match time/date

/em/{id}-list.json — Match Events


```
[
  {
    "half": 1,
    "time": 9,
    "action": "КК",
    "number1": "74",
    "first_name1": "Рафаэль",
    "last_name1": "Саксесс",
    "team1": "Жетісу",
    "number2": "",
    "first_name2": "",
    "last_name2": "",
    "team2": "",
    "standard": null
  },
  {
    "half": 2,
    "time": 88,
    "action": "ГОЛ",
    "number1": "79",
    "first_name1": "Глеб",
    "last_name1": "Валгушев",
    "team1": "Қызылжар",
    "number2": "31",
    "first_name2": "Михаил",
    "last_name2": "Голубничий",
    "team2": "Жетісу",
    "standard": null
  }
]
```


Action types → QFL event_type mapping:

sota action	QFL event_type	Notes
ГОЛ	goal	player2 = **opponent** (GK), NOT assist
АВТОГОЛ	own_goal	
ПЕНАЛЬТИ	penalty	Scored penalty
НЕЗАБИТЫЙ ПЕНАЛЬТИ	missed_penalty	
ГОЛЕВОЙ ПАС	assist	Separate event, linked to goal by (half, minute, team_id)
ЖК	yellow_card	
2ЖК	second_yellow	

КК	red_card	
ЗАМЕНА	substitution	player1=off, player2=on, same team1/team2

> **Critical:** On ГОЛ events, player2 is the **goalkeeper/opponent who conceded**, NOT the assisting player. Assists come as separate ГОЛЕВОЙ ПАС events and are linked by (half, minute, team_id).

Deduplication key: (half, minute, event_type, player_id) or (half, minute, event_type, normalized_name)

/em/{id}-stat.json — Live Match Stats (109 metrics)

```
[
  {"metric": "name",          "home": "Жетісу",  "away": "Қызылжар"},
  {"metric": "goals",         "home": 0,         "away": 1},
  {"metric": "goals_1",       "home": 0,         "away": 0},
  {"metric": "goals_2",       "home": 0,         "away": 1},
  {"metric": "shots",         "home": 9,         "away": 17},
  {"metric": "shots_on_target","home": 4,        "away": 3},
  {"metric": "corners",       "home": 4,         "away": 3},
  {"metric": "fouls",         "home": 14,        "away": 17},
  {"metric": "yc",            "home": 1,         "away": 6},
  {"metric": "rc",            "home": 1,         "away": 0},
  {"metric": "possessions",   "home": "48%",     "away": "52%"},
  {"metric": "saves",         "home": 1,         "away": 4}
]
```

Each metric has halves variants: goals, goals_1, goals_2, … goals_5 (for extra time). possessions is a **string** "48%", not a number.

/em/{id}-referies.json

```
[
  {"kind": "MAIN",                 "name": "Иван Иванов"},
  {"kind": "ASSISTANT1",           "name": "Петр Петров"},
  {"kind": "ASSISTANT2",           "name": "Сергей Сергеев"},
  {"kind": "REFEREE4",             "name": "Алексей Алексеев"},
  {"kind": "ASSISTANT_VIDEO_MAIN", "name": "Дмитрий Дмитриев"}
]
```

/em/{id}-lineups-home.json / -lineups-away.json — Simple Lineup

```
[
  {"number": "Жеңіс",     "full_name": "away"},
  {"number": "СТАРТОВЫЙ","full_name": ""},
  {"number": 1,           "full_name": "Сергей Игнатович"},
  {"number": 4,           "full_name": "Саги Совет"}
]
```

Simpler format (just number + name). Uses СТАРТОВЫЙ section marker (vs ОСНОВНЫЕ in team-home.json).

/em/{id}-players-home.json / -players-away.json — Player Stats

```
[
  {
    "kind": "home",
    "team": "Кайсар",
    "number": 7,
```

```
    "first_name": "Елжас",
    "last_name": "Алтынбеков",
    "full_name": "Елжас Алтынбеков",
    "id": "f6d93df2-3a63-48ec-8c1e-3ecef5386547",
    "goals": 0,
    "assists": 0,
    "yc": 0,
    "rc": 0
  }
]
```

Compact stats: goals, assists, yellow_cards (`yc`), red_cards (`rc`). Useful for real-time scoreboard.

---

4. Use Case 1 — Reference Data Sync

Schedule: `sync-references-daily` — 06:00 Asia/Almaty **Service:** `ReferenceSyncService`

Sync Order

```
1. Tournaments (no deps)
2. Seasons (3 languages: RU, KZ, EN in parallel)
3. Teams (per season_id in SYNC_SEASON_IDS)
```

Tournament → Championship Mapping

sota.id has no concept of "championship" — QFL maps via `Championship.sota_ids` (array of tournament IDs stored in the DB). The reverse lookup happens in `ReferenceSyncService`.

Season Sync (3 languages)

For each language, seasons are fetched with `Accept-Language` header. Results are merged:

```
# Pseudo-code
ru_seasons = await client.get_seasons(lang="ru")
kz_seasons = await client.get_seasons(lang="kk")
en_seasons = await client.get_seasons(lang="en")
# Merge by id → {id: {name_ru: ..., name_kz: ..., name_en: ...}}
```

Team Photo Sync

Team logos from sota.id are fetched and stored in MinIO:

```
sota.id URL → MinIO object name → DB stores object name → FileUrlType resolves on read
```

---

5. Use Case 2 — Player Sync

Schedule: part of `sync-games-every-2h` **Service:** `PlayerSyncService`

Player Deduplication

Players are identified by `Player.sota_id` (UUID from sota.id `id` field). A player is auto-created if they appear in stats but don't exist in the DB:

```
async def _get_or_create_player_by_sota(self, sota_uuid: str, ...) -> int:
    # 1. Try lookup by sota_id
    # 2. Try lookup by name + team
    # 3. Create new player
```

Player Season Stats

`/api/public/v2/players/{id}/season_stats/?season_id=X` returns `[{key, value, name}]`.

All 50+ fields are mapped to `PlayerseasonStats` columns. Fields not in `PLAYER_SEASON_STATS_FIELDS` go to `extra_stats` (JSONB).

Key stats fields (from `base.py`): - Basic: `games_played`, `games_starting`, `time_on_field_total` - Goals: `goal`, `goal_pass`, `goal_and_assist`, `owngoal`, `penalty_success` - Shots: `shot`, `shots_on_goal`, `shots_blocked_opponent` - Passes: `pass`, `pass_ratio`, `key_pass`, `pass_cross`, `pass_progressive` - Duels: `duel`, `duel_success`, `aerial_duel`, `ground_duel` - Defense: `tackle`, `interception`, `recovery` - Discipline: `yellow_cards`, `second_yellow_cards`, `red_cards` - GK: `save_shot`, `goals_conceded`, `dry_match` - Advanced: `xg`, `xg_per_90`

---

6. Use Case 3 — Game Stats Sync

Schedule: `sync-live-stats-every-15min` **Service:** `GameSyncService`

Prerequisite

A game **must have `sota_id` set** (UUID in `Game.sota_id`) for stats sync to work. If `sota_id` is null, sync is skipped.

Game Stats Flow

```
sync_game_stats(game_id)
  ├── get_game_player_stats(sota_id)  → /games/{id}/players/
  │      └── upsert PlayerGameStats rows
  └── get_game_team_stats(sota_id)    → /games/{id}/teams/
         └── upsert TeamGameStats rows
```

Fields stored per player per game (`GAME_PLAYER_STATS_FIELDS`): `goals`, `assists`, `shot`, `shots_on_goal`, `shots_off_goal`, `pass`, `pass_accuracy`, `duel`, `tackle`, `corner`, `offside`, `foul`, `yellow_cards`, `red_cards`

Fields stored per team per game (`GAME_TEAM_STATS_FIELDS`): `possession`, `possession_percent`, `shot`, `shots_on_goal`, `shots_off_goal`, `pass`, `pass_accuracy`, `foul`, `yellow_cards`, `red_cards`, `corner`, `offside`

Game Events Flow

```
sync_game_events(game_id)
  └── get_live_match_events(sota_id)  → /em/{sota_id}-list.json
         └── parse + deduplicate → insert GameEvent rows
```

Assist linking algorithm:

```
For each event in list:
```

```
  if action == "ГОЛЕВОЙ ПАС":
    # Find a goal in same half, within ±1 minute, by same team
    # Link as assist_game_event_id → goal_event_id
```

Deduplication: `UNIQUE(game_id, half, minute, event_type, player_id)` + fallback on normalized player name.

---

7. Use Case 4 — Season Statistics

Schedule: part of `sync-games-every-2h` **Services:** `StatsSyncService`

Score Table

```
sync_score_table(season_id)
  └── GET /seasons/{id}/score_table/
        └── goals string "53:19" → goals_scored=53, goals_conceded=19
```

Team Season Stats (92 metrics)

```
sync_team_season_stats(season_id)
  └── For each team in season:
        GET /v2/teams/{id}/season_stats/?season_id=X
        → [{key, value, name}] → upsert TeamSeasonStats
```

All 92 fields are in `TEAM_SEASON_STATS_FIELDS`. Extra fields → `extra_stats` JSONB.

---

8. Use Case 5 — Pre-Game Lineup

Triggered by: `check-upcoming-games-every-5min` → `sync_pre_game_lineup(game_id)` **Service:** `LineupSyncService`

Sync Flow

```
sync_pre_game_lineup(game_id)
  └── GET /games/{sota_id}/pre_game_lineup/
        ├── Upsert referees (main, assistants, 4th, VAR, inspector)
        ├── Upsert coaches (home + away)
        └── Upsert lineup players (GameLineup rows)
              └── Player lookup: by sota_id → by name → create
```

Lineup Response Structure

```
{
  "data": {
    "referee": {"main": "Name", "1st_assistant": "...", "2nd_assistant": "...", "4th_ref
    "home_team": {
      "coach": "Name",
      "lineup": [{"player_id": "uuid", "first_name": "...", "last_name": "...", "player_
    },
    "away_team": {"...": "same structure"}
  }
```

```
}
```

9. Use Case 6 — Live Match Data

Schedule: `sync-live-events-every-5sec` + `check-upcoming-games-every-5min` **Service:** `LineupSyncService` + `GameSyncService`

Live Data Sources (priority order)

```
1. sota.id /em/{sota_id}-team-{side}.json       → positions, kits, formation
2. vsporte.ru /api/v2/qfl/files/{vsporte_id}_team_{side}.json  → fallback
```

Live Lineup Processing (`sync_live_positions_and_kits`)

```
for side in ["home", "away"]:
    live_data = await _fetch_from_sota(game.sota_id, side)
    # OR
    live_data = await _fetch_from_vsporte(game.vsporte_id, "host"/"guest")

    for entry in live_data:
        # Extract marker rows → formation, coach, kit_color
        # Extract player rows → update amplua, field_position, lineup_type
```

Lineup Entry Parsing (`lineup_feed_parser.py`)

The `normalize_lineup_entry()` function handles both sota.id and VSporte formats:

```
STARTING_MARKERS = {"ОСНОВНЫЕ", "STARTING"}
SUBS_MARKERS = {"ЗАПАСНЫЕ", "SUBS"}
COACH_MARKERS = {"COACH", "MAIN"}
FORMATION_MARKER = "FORMATION"
TEAM_MARKER = "TEAM"
```

- Entries with `number` in `STARTING_MARKERS` → set `current_section = starter`
- Entries with `number` in `SUBS_MARKERS` → set `current_section = substitute`
- Integer `number` entries → actual player row with `amplua` and `position`
- `FORMATION` entry → extract formation string and kit color `#RRGGBB` from `full_name`

VSporte Fallback

```
URL: https://broadcast.vsporte.ru/api/v2/qfl/files/{vsporte_id}_team_{side}.json
Side mapping: "home" → "host", "away" → "guest"
```

VSporte uses `vsporte_id` stored in `Game.vsporte_id`. The response format is identical to sota.id `/em/` lineup format (same parser handles both).

Live Caching

```
LINEUP_LIVE_REFRESH_TTL_SECONDS = 30    # In-memory cache TTL
```

```
LINEUP_LIVE_REFRESH_TIMEOUT_SECONDS = 3  # HTTP request timeout
```

10. VMix / TV Broadcast Endpoints

These endpoints are designed for video production (OBS/vMix broadcast software). They return data in a TV-friendly flat format with BAS file paths.

Generate VMix Links

```
GET /api/public/v1/games/{id}/vmix/generate_link/
```

Returns all available stream links including undocumented `/em/` endpoints:

```
[
  {"name": "team-home",      "link": "https://sota.id/em/{id}-team-home.json?access_toke
  {"name": "team-away",      "link": "https://sota.id/em/{id}-team-away.json?access_toke
  {"name": "list",           "link": "https://sota.id/em/{id}-list.json?access_token=...
  {"name": "stat",           "link": "https://sota.id/em/{id}-stat.json?access_token=...
  {"name": "referies",       "link": "https://sota.id/em/{id}-referies.json?access_token
  {"name": "lineups-home",   "link": "https://sota.id/em/{id}-lineups-home.json?access_t
  {"name": "lineups-away",   "link": "https://sota.id/em/{id}-lineups-away.json?access_t
  {"name": "players-home",   "link": "https://sota.id/em/{id}-players-home.json?access_t
  {"name": "players-away",   "link": "https://sota.id/em/{id}-players-away.json?access_t
]
```

Best Players in Game (vmix)

```
GET /api/public/v1/games/vmix/{id}/best-players/{team_id}/?metric=shot
```

Required: `metric` param (400 if missing).

```
[
  {"position": "Позиция", "metric_en": "Название метрики", "...": "header row"},
  {
    "position": "",
    "metric_en": "Shots",
    "metric_kz": "Соққылар",
    "image_bas": "C:\\KPL_FINAL\\SOTA_PHOTO_PLAYER\\Kaisar\\GUBAREV_NIKITA.png",
    "name_en": "Nikita Gubarev",
    "name_kz": "Никита Губарев",
    "value": 5,
    "main_team_en": "Kaysar"
  }
]
```

All Player Stats Matrix (vmix v2)

```
GET /api/public/v2/games/vmix/{id}/team-players-stats/{team_id}/
```

Returns a matrix where rows are metrics and columns are player names:

```
[
  {"metric_kz": "Название метрики", "Елжас Алтынбеков": "Елжас Алтынбеков", "...": "all p
  {"metric_kz": "Ссылка на лого команды", "Елжас Алтынбеков": "C:\\KPL_FINAL\\LOGO_KOMAN
  {"metric_kz": "Ссылка на фото", "Елжас Алтынбеков": "C:\\KPL_FINAL\\...\\ALTYNBEKOV.pn
  {"metric_kz": "Соққылар", "Елжас Алтынбеков": 3, "Нурмат Сарсенов": 1, "...": "per-pla
]
```


Last 5 Games (vmix)

```
GET /api/public/v1/seasons/vmix/{season_id}/last-games/{team_id}/
```


```
[
  {
    "opponent_en": "Jenis",
    "opponent_kz": "Жеңіс",
    "opponent_image_bas": "C:\\KPL_FINAL\\LOGO_KOMAND\\Jenis.png",
    "score": "2-2",
    "date": "26.10",
    "result": "#F49C00",
    "main_team_en": "Kaysar"
  }
]
```


`result` is a HEX color: `#1DB954` = win, `#F49C00` = draw, `#E0032B` = loss.

Season Score Table (vmix)

```
GET /api/public/v1/seasons/vmix/{season_id}/score_table/
```


```
[
  {
    "position": 1,
    "name_kz": "Қайрат",
    "image_bas": "C:\\KPL_FINAL\\LOGO_KOMAND\\Kairat.png",
    "rg": 34,
    "wins": 18,
    "draws": 5,
    "losses": 3,
    "goals": "53:19",
    "points": 59,
    "matches": 26
  }
]
```


Teams Season Stats Ranking (vmix)

```
GET /api/public/v1/seasons/vmix/{season_id}/teams_season_stats/?metric=shot
```


```
[
  {"position": "Позиция", "...": "header row"},
  {
    "metric_en": "Shots",
```

```
    "metric_kz": "Соққылар",
    "image_bas": "C:\\KPL_FINAL\\LOGO_KOMAND\\Kairat.png",
    "name_en": "Kairat",
    "name_kz": "Қайрат",
    "value": 448,
    "secondary_value": 17.2,
    "games": 26,
    "position": 1
  }
]
```

Two-Team Average Stats Comparison (vmix)

```
GET /api/public/v1/seasons/vmix/{season_id}/teams_average_season_stats/{game_id}/
```

Returns 11 rows comparing two teams playing in the specified game (season averages):

```
[
  {"metric_en": "name",                    "team_home": "Kaysar",  "team_away": "Jenis"},
  {"metric_en": "logo",                    "team_home": "C:\\...\\Kaisar.png", "team_away
  {"metric_en": "Possessions per match",  "team_home": 18.7,      "team_away": 29.2},
  {"metric_en": "Goals per match",        "team_home": 0.9,       "team_away": 1.3},
  {"metric_en": "Yellow cards per match", "team_home": 1.7,       "team_away": 2.1},
  {"metric_en": "Red cards per match",    "team_home": 0.1,       "team_away": 0.1},
  {"metric_en": "Accurate passes per match","team_home": 219.3,   "team_away": 454.3},
  {"metric_en": "Corner per match",       "team_home": 3.4,       "team_away": 3.7}
]
```

Team's Best Players for Season (vmix)

```
GET /api/public/v1/seasons/vmix/{season_id}/teams/{team_id}/best-players/{game_id}/?metr
```

```
[
  {"position": "Позиция", "...": "header row"},
  {
    "position": "",
    "metric_en": "Shots",
    "metric_kz": "Соққылар",
    "image_bas": "C:\\...\\ABIKEN_AIBOL.png",
    "name_en": "Aybol Abiken",
    "name_kz": "Айбол Әбікен",
    "value": "45",
    "main_team_en": "Kaysar"
  }
]
```

11. Celery Tasks & Scheduling

```
# backend/app/tasks/__init__.py
# Active only when SOTA_ENABLED=True
```

Task ID	Schedule	Action
`sync-references-daily`	06:00 Asia/Almaty	Tournaments + seasons (3 lang) + teams
`sync-games-every-2h`	`0 */2 * * *`	Players + game data + player stats
`sync-live-stats-every-15min`	`*/15 * * * *`	Game stats for recent/live games
`check-upcoming-games-every-5min`	`*/5 * * * *`	Detect upcoming games, pre-game lineup
`sync-live-events-every-5sec`	every 5.0s	Live events from `/em/-list.json`
`end-finished-games-every-10min`	`*/10 * * * *`	Mark finished live games

Feature Flags

Flag	Effect
`SOTA_ENABLED=false`	`beat_schedule = {}` — disables ALL Celery sync
`Season.sync_enabled=False`	Skip all sync operations for that season
`Game.sync_disabled=True`	Skip sync for that specific game

12. Environment Variables

Variable	Default	Description
`SOTA_ENABLED`	`true`	Master switch for Celery sync pipeline
`SOTA_API_EMAIL`	—	Email for JWT authentication
`SOTA_API_PASSWORD`	—	Password for JWT authentication
`SOTA_API_BASE_URL`	`https://sota.id/api`	API base URL
`SYNC_SEASON_IDS`	`[61,85,71,80,84]`	Season IDs synced by Celery tasks. **Add 173 for KPL 2026**
`LINEUP_LIVE_REFRESH_TTL_SECONDS`	`30`	In-memory cache TTL for live lineups
`LINEUP_LIVE_REFRESH_TIMEOUT_SECONDS`	`3`	HTTP timeout for live lineup requests

13. Unused Endpoints — Development Potential

Endpoint	Priority	What can be built	Data availability
`GET /seasons/{id}/best_players/?metric=goal&max=10&tour=N`	☆☆☆ HIGH	Scorers table, assists table, MVP widget	Active season only — data cleared after end
`GET /seasons/{id}/team_of_week/?tour=N`	☆☆☆ HIGH	"Team of the Tour" page / widget	Active season only (without tour: last 3 tours)
`GET /v2/players/{id}/game_stats/?game_id=X`	☆☆☆ HIGH	Detailed player profile per game (50 metrics)	Works for all seasons
`GET /v1/teams/{id}/season_stats_v2/?tour=N&single_tour=bool`	☆☆ MED	Team performance trend by tour (92 metrics)	Works for all seasons

`GET /seasons/{id}/season_stats_v2/`	☆☆ MED	League-wide aggregate stats (71 metrics)	Works for all seasons
`GET /v2/seasons/{id}/season_stats/`	☆☆ MED	Same 71 metrics (v2 alt)	Works for all seasons
`GET /metrics/`	☆ LOW	Dynamic metric catalog (293 metrics, 287 unique keys)	Always available
`GET /em/{id}-players-{side}.json`	☆☆ MED	78 fields per player with per-half breakdown	During/after match
`GET /vmix/.../teams_average_season_stats/{game_id}/`	☆☆ MED	Pre-match two-team comparison widget	Works for all seasons
All other `vmix` endpoints	☆ LOW	TV broadcast / OBS integration	Varies

Implementing `best_players`

```
# New SotaClient method:
async def get_best_players(
    self, season_id: int, metric: str, max: int = 10, tour: int | None = None
) -> dict:
    params = {"metric": metric, "max": max}
    if tour:
        params["tour"] = tour
    response = await self._make_request(
        "get",
        f"{self.base_url}/public/v1/seasons/{season_id}/best_players/",
        params=params,
    )
    return response.json()
```

Implementing `team_of_week`

```
async def get_team_of_week(self, season_id: int, tour: int | None = None) -> dict:
    params = {"tour": tour} if tour else {}
    response = await self._make_request(
        "get",
        f"{self.base_url}/public/v1/seasons/{season_id}/team_of_week/",
        params=params,
    )
    return response.json()
```

14. Manual Operations

Re-sync Assists

```
# Single game
docker exec qfl-backend python3 scripts/resync_assists.py <game_id>

# All games
docker exec qfl-backend python3 scripts/resync_assists.py --all

# All games in season
```

```
docker exec qfl-backend python3 scripts/resync_assists.py --season <season_id>
```

Use when assist linking fails (team name mismatch, minute offset).

Trigger Sync via Admin API

```
# Trigger reference sync
curl -X POST https://kffleague.kz/api/v1/admin/sync/references/

# Trigger game sync for season
curl -X POST https://kffleague.kz/api/v1/admin/sync/games/?season_id=61

# Sync specific game events
curl -X POST https://kffleague.kz/api/v1/admin/sync/game-events/<game_id>/
```

---

15. Troubleshooting

Problem	Cause	Fix
Stats not syncing for a game	`Game.sota_id = NULL`	Set `sota_id` via Admin → Games
Assists not linked	Team name mismatch or ±1min offset	Run `scripts/resync_assists.py <game_id>`
Sync skipped for season	`Season.sync_enabled = False`	Enable in Admin → Seasons
Player not found in lineup	`Player.sota_id` not set	Player auto-created; check for duplicates
`401 Unauthorized` on API	Expired JWT or wrong credentials	Check `SOTA_API_EMAIL`/`SOTA_API_PASSWORD` in `.env`, restart Celery
`404` on `/em/` endpoint	Game not started or wrong sota_id	Expected for future games; verify `Game.sota_id`
Duplicate events	Re-sync triggered twice	Deduplication runs automatically; safe to re-sync
`score_table_v2` returns null	v2 endpoints not available for some seasons	Use v1: `/seasons/{id}/score_table/`
`best_players` returns empty	Season completed without tour param	Pass explicit `tour=N` for historical data
Live lineup shows wrong kits	Colors come from FORMATION marker	Re-run `sync_live_positions_and_kits`
VSporte fallback not working	`Game.vsporte_id = NULL`	Set vsporte_id in Admin or use sota_id directly